

MODULE 2.1: HACKERS & CYBER CRIMES

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Hacking

- **Hacker** — A person skilled in computers/networks who explores, tests, and finds weaknesses. *Not inherently malicious*; can be ethical (security researchers) or malicious.
- **Cracker** — Someone who breaks into systems or circumvents protections with *malicious intent* (stealing, vandalism, or disruption).
- **Practical note:** Modern usage often calls malicious actors “hackers,” but academically it’s useful to distinguish *hackers = skill/curiosity* vs *crackers = malicious breakers*.
- **Hacking:** the act of probing, exploiting, or manipulating computer systems, networks, or applications to discover weaknesses or achieve an objective (could be learning, research, theft, or disruption).

Classification of Hackers

1. Based on Intent (Ethical Stance)

1. **White-hat hackers** are ethical hackers who use their skills to find vulnerabilities in systems, networks, or applications legally and responsibly. They help organizations strengthen security by conducting penetration testing and reporting their findings to prevent malicious exploitation.
2. **Grey-hat hackers** operate in a legal and ethical gray zone. They may explore systems without explicit permission but do not have purely malicious intentions. Often, they disclose discovered vulnerabilities either to the system owner or publicly, sometimes causing minor disruption unintentionally.
3. **Black-hat hackers** have malicious intent and exploit vulnerabilities for personal gain, financial profit, or to cause harm. They break into systems without authorization and often steal data, deploy malware, or disrupt operations.
4. **Blue-hat hackers** are security professionals who are invited externally to find vulnerabilities before software is released. They focus on testing applications and providing feedback to developers to prevent attacks once the software is publicly available.

2. Based on Experience or Skill Level

1. **Novices or beginners/Script-kiddies** are individuals who have very basic knowledge of computer systems and networks. They typically use automated tools or tutorials to exploit vulnerabilities but lack the ability to develop custom attacks.
2. **Intermediate hackers** understand networks, protocols, and common vulnerabilities. They can perform basic attacks, modify existing exploits, and use scripting to automate certain hacking tasks.
3. **Advanced hackers** have in-depth knowledge of operating systems, networking, programming, and security vulnerabilities. They can develop custom exploits, reverse-engineer software, and evade detection mechanisms.
4. **Expert or elite hackers** possess exceptional skills and deep technical knowledge. They can discover zero-day vulnerabilities, building sophisticated malware, conducting advanced persistent threats (APT), and orchestrating complex, stealthy attacks on high-value targets.

3. Based on Motivation

1. **Financial Gain:** Hackers motivated by money target financial institutions, e-commerce platforms, or personal accounts. They may steal money, commit fraud, deploy ransomware, or sell stolen credentials for profit.
2. **Political / Nation-State Objectives:** These hackers work for governments or nation-states. They perform espionage, sabotage, or cyberwarfare to gain strategic advantage, such as stealing defense data or disrupting infrastructure.
3. **Ideological / Social Causes (Hacktivism):** Hacktivists target organizations or governments to advance political, social, or environmental causes. They may deface websites, leak information, or conduct DDoS attacks to send a message.
4. **Personal Grudge / Revenge:** Attacks motivated by personal resentment come from ex-employees, competitors, or individuals seeking revenge. They aim to damage systems, steal data, or disrupt operations.
5. **Curiosity / Reputation:** Some hackers act to explore systems, learn vulnerabilities, or gain recognition. Though not always malicious, their actions can still cause disruption or accidental damage.
6. **Corporate Espionage:** Hackers target companies to steal trade secrets, intellectual property, or sensitive business information. The goal is competitive advantage or financial gain.

4. Based on Profile / Persona

1. **Novices:** These are beginners with limited skills who often rely on pre-written scripts or tools to hack systems. They are mostly curious or want recognition, but lack deep technical knowledge.
2. **Cyber-punks:** Hackers motivated by rebellion, anti-establishment ideologies, or thrill-seeking. They often target corporations or governments to protest, create disruption, or challenge authority.
3. **Internals:** Insiders who already have authorized access to systems, such as employees, contractors, or partners. They exploit their legitimate access to steal data, manipulate systems, or cause damage.
4. **Old Guard Hackers:** Experienced hackers from earlier generations who have deep technical knowledge and a historical understanding of hacking methods. They may act as mentors, security researchers, or underground operators.

5. **Coders:** Highly skilled programmers who create hacking tools, malware, or exploits. Their expertise allows them to develop custom attacks and advanced software for penetration testing or malicious purposes.
6. **Professional Criminals:** Hackers who engage in organized cybercrime for monetary gain. They run ransomware campaigns, phishing operations, identity theft, or financial fraud professionally, often as part of criminal networks.
7. **Information Warriors / Terrorists:** Hackers targeting national or organizational infrastructure to create fear, sabotage critical systems, or advance geopolitical agendas. Often state-sponsored or part of terrorist organizations.
8. **Hactivists:** Individuals or groups using hacking to promote social, political, or environmental causes. They aim to raise awareness, protest policies, or disrupt organizations symbolically rather than for personal profit.

Active Attacks

Active attacks are cyber attacks where the attacker intentionally modifies, disrupts, damages, or destroys the system, network, or data. The purpose is to directly affect the confidentiality, integrity, or availability (CIA triad) of the target. Unlike passive attacks, active attacks are detectable because they cause noticeable changes or interruptions.

1. **Intentional and Malicious:** The attacker deliberately targets the system to cause harm or unauthorized changes.
2. **Data Modification:** Active attacks often involve altering messages, databases, or files.
3. **Service Disruption:** Many active attacks aim to disrupt normal operations of a system or network.
4. **Easier to Detect:** Because active attacks modify or damage resources, they are more likely to trigger alarms or system alerts.
5. **Risk to Assets:** Targets tangible and intangible assets, such as computers, databases, intellectual property, and confidential information.
6. Active attacks **always involve interaction with the target system** that changes its state.
7. They **pose immediate risks** to the confidentiality, integrity, and availability of assets.
8. **Countermeasures** include firewalls, intrusion detection and prevention systems (IDS/IPS), strong access controls, monitoring, patch management, and incident response protocols.

Types / Examples of Active Attacks

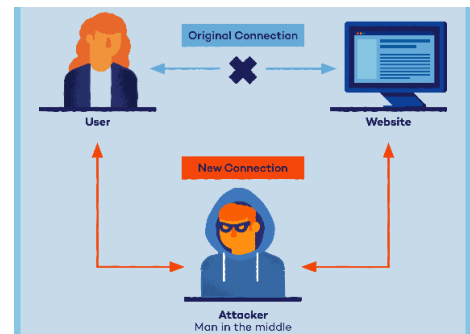
1. **Malware Attacks :** Malicious software designed to disrupt, damage, or gain unauthorized access to systems.
 - **Examples:** Viruses, worms, Trojans, ransomware, spyware (if it modifies files or systems).
 - **Impact:** Corrupts or encrypts data, crashes systems, or disrupts operations.
2. **Denial of Service (DoS) / Distributed DoS (DDoS):** Overwhelming a system or network with traffic to make it unavailable for legitimate users.
 - **Examples:** Flooding a website with requests, botnet-driven DDoS attacks.
 - **Impact:** Interrupts business services, causes revenue loss, damages reputation.

DoS Attack



3. **Man-in-the-Middle (MitM) Attacks (Active Mode):** Intercepting communication to **alter or inject malicious data** between two parties.

- **Examples:** Modifying messages in transit, injecting malware via intercepted connections.
- **Impact:** Data integrity is compromised, financial or confidential data can be stolen or manipulated.



4. **SQL Injection and Web Attacks :** Exploiting vulnerabilities in web applications to **manipulate databases or website functions**.

- **Examples:** Changing database records, bypassing authentication, defacing websites.
- **Impact:** Data loss, unauthorized access, service disruption.



5. **Password Attacks:** Actively attempting to gain unauthorized access by cracking or guessing passwords.

- **Examples:** Brute-force attacks, dictionary attacks, credential stuffing.
- **Impact:** Unauthorized access to accounts, potential data theft or modification.

6. **Advanced Persistent Threats (APT) :** Long-term, targeted attacks by skilled attackers to infiltrate and **maintain access** for espionage or sabotage.

- **Examples:** State-sponsored malware campaigns, prolonged system compromises.
- **Impact:** Stealthy data exfiltration, operational disruption, strategic information theft.

7. **Insider Threats:** Authorized users misusing their access to perform malicious actions.

- **Examples:** Employees deleting critical files, stealing data, or modifying sensitive systems.
- **Impact:** Can be severe because insiders already have access; may lead to financial, operational, or reputational damage.

8. **Spoofing Attacks :** An attack where an attacker **forges or impersonates identity information** to appear as a trusted source in order to gain unauthorized access or manipulate communication.

- **Examples:** IP spoofing, ARP spoofing, DNS spoofing/cache poisoning, email/sender spoofing.
- **Impact:** Enables man-in-the-middle attacks, session hijacking, traffic redirection, credential theft, and unauthorized access to systems or data



Passive Attacks

A passive attack is one where an adversary observes, monitors, or captures information from a system or network without altering the data, system state, or service. The attacker's goal is usually to gather intelligence (confidential information) quietly and stealthily.

- **Non-disruptive:** Does not change or damage data or services, so it's often hard to detect.
- **Targets confidentiality:** Primarily compromises secrecy (not integrity or availability).

- **Stealthy & reconnaissance-driven:** Frequently used to map a target, steal credentials, or learn system behavior before launching active attacks.
- **Low noise:** Generates little to no observable anomalies in logs or performance.

Common Types & Examples

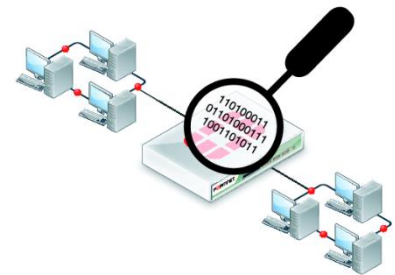
1. Phishing Attacks

- **Definition:** An attack where the attacker **tricks users into revealing sensitive information** such as usernames, passwords, or financial details by masquerading as a legitimate source.
- **Examples:** Email phishing, spear phishing, whaling, fake websites.
- **Impact:** Credential theft, unauthorized access to accounts, identity theft, and financial loss.



2. Network Sniffing / Packet Capture

- **Definition:** An attack where the attacker **monitors and captures network traffic** to intercept sensitive information.
- **Examples:** Using tools like Wireshark or tcpdump to capture packets on LANs or Wi-Fi networks.
- **Impact:** Exposure of passwords, session tokens, confidential data, and potential preparation for active attacks.

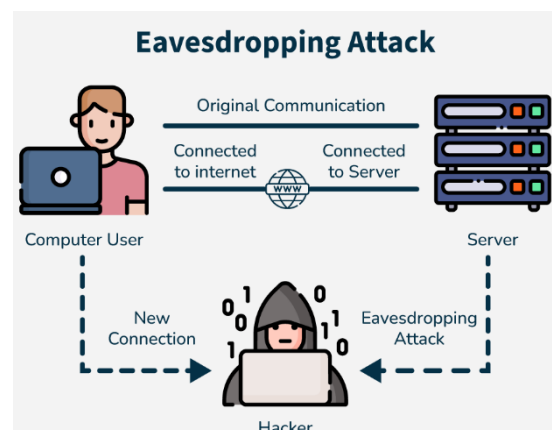


3. Traffic Analysis

- **Definition:** Observing patterns, timing, and volume of network traffic to **infer sensitive information** without directly accessing the content.
- **Examples:** Monitoring peak usage times, volume of encrypted messages, communication patterns between systems.
- **Impact:** Leakage of system behavior, relationships, or operational secrets, often used to plan active attacks.

4. Eavesdropping

- **Definition:** Listening in on communications to **capture sensitive information** without altering it.
- **Examples:** Intercepting phone calls, VoIP traffic, emails, or instant messages.
- **Impact:** Confidentiality breach, theft of personal or corporate information.



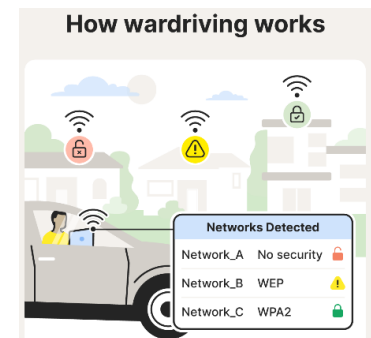
5. Shoulder Surfing

- **Definition:** Observing a person directly to **obtain sensitive information** such as passwords or PINs.
- **Examples:** Watching someone enter credentials at an ATM or login screen.
- **Impact:** Unauthorized access, identity theft, or financial fraud.



6. Wardriving

- **Definition:** Searching for and mapping **wireless networks** from a moving vehicle to find insecure access points.
- **Examples:** Using laptops or mobile devices with Wi-Fi scanners to identify open networks.
- **Impact:** Exposure of network locations, potential for unauthorized access, and planning of further attacks.



7. Passive Man-in-the-Middle (MitM)

- **Definition:** Intercepting communication between two parties **without altering data** to eavesdrop on sensitive information.
- **Examples:** Sniffing encrypted traffic or monitoring session tokens in transit.
- **Impact:** Breach of confidentiality and potential preparation for future active attacks.

8. Passive Reconnaissance / Footprinting

- **Definition:** Gathering publicly available information about a target **without interacting with it directly**.
- **Examples:** WHOIS lookups, public DNS records, company websites, social media.
- **Impact:** Collects intelligence for planning active attacks; compromises operational secrecy.

9. Electromagnetic / Side-Channel Monitoring

- **Definition:** Observing **indirect emissions** such as power usage, electromagnetic radiation, or timing to infer sensitive information.
- **Examples:** TEMPEST attacks, measuring CPU power patterns, or monitoring keyboard signals.
- **Impact:** Unauthorized access to confidential data without touching the system; often used to extract encryption keys or sensitive computations.

Impact

- **Confidentiality breach:** Exposure of credentials, personal data, trade secrets.
- **Enabler for active attacks:** Information gathered passively (e.g., credentials or network maps) often enables later active attacks like session hijacking, spoofing, or targeted exploits.
- **Difficult to notice:** Long periods of undetected intelligence collection can cause large-scale harm when leveraged.

Detection

- **Hard to detect directly:** Because passive attacks don't change packets or disrupt service, traditional IDS/IPS often miss them.

- **Indirect indicators:** Unusual device in promiscuous mode, unexplained network taps, unexpected increases in outbound data, or anomalous timing/pattern changes can be clues.

Aspect	Active Attacks	Passive Attacks
Definition	Attacks where the attacker alters, disrupts, damages, or manipulates the system, data, or network.	Attacks where the attacker observes, monitors, or captures information without changing the system or data.
Goal / Purpose	To modify, destroy, or disrupt systems and data; impact confidentiality, integrity, and availability.	To gather intelligence or sensitive information stealthily; mainly affects confidentiality.
Detectability	Easier to detect due to changes or disruption in data, service, or system behavior.	Harder to detect because it does not alter data or system behavior.
Impact on CIA Triad	Compromises Integrity, Availability, and sometimes Confidentiality .	Primarily compromises Confidentiality .
Examples	Malware (viruses, worms, ransomware), Spoofing, DoS/DDoS, SQL Injection, Password Attacks, Active MitM, Insider Threats, APTs.	Network sniffing / packet capture, Traffic Analysis, Eavesdropping, Shoulder Surfing, Wardriving, Passive MitM, Passive Reconnaissance, Electromagnetic / Side-Channel Monitoring, Phishing (passive observation type).
Nature of Attack	Interactive: attacker manipulates or injects data or traffic.	Non-interactive / Observation: attacker only monitors.
Risk Level	Immediate, noticeable damage; can disrupt operations.	Stealthy, long-term risk; may enable future active attacks.
Countermeasures	Firewalls, IDS/IPS, patching, strong authentication, monitoring, incident response, encryption for modified data.	Encryption (TLS/HTTPS), strong authentication, VPNs, network monitoring, physical security, traffic obfuscation, reducing information leakage.

Which one is more dangerous and why ?

1. Active Attacks – Usually More Immediately Dangerous

- **Why:** They **directly alter, destroy, or disrupt** systems, data, or services.
- **Impact:** Can crash servers, corrupt databases, hijack sessions, or cause service outages.
- **Examples:** Malware (viruses, ransomware), DoS/DDoS, spoofing, SQL injection, Active MitM.
- **Detection:** Easier to detect because they leave traces in logs, trigger alerts, or affect system behavior.
- **Danger Level:** High **short-term damage**; can also compromise integrity and availability in addition to confidentiality.

2. Passive Attacks – Stealthy but Long-Term Risk

- **Why:** They **don't change data or disrupt services**, so they are often **undetected for long periods**.
- **Impact:** Primarily leaks sensitive information, enabling attackers to plan **future active attacks**.
- **Examples:** Sniffing, traffic analysis, eavesdropping, passive MitM, footprinting.
- **Detection:** Very difficult to detect because they leave minimal traces.
- **Danger Level:** High **long-term risk**, mainly to confidentiality; may indirectly lead to more severe attacks.

Verdict

- **Immediate danger:** Active attacks are more destructive **right away**.
- **Strategic danger:** Passive attacks are dangerous **silently** because they can prepare the ground for future destructive attacks.
- **Overall:** Both are serious, but **active attacks are more visibly damaging**, while **passive attacks are stealthily enabling and dangerous over time**.